IRCA Exposure and Composition of Admissions

How denominator choice changes a headline admissions-composition metric

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February 2026

Executive Summary

This project builds a reproducible country–year panel of 19 Latin American origin countries from 1982 to 2000 to study whether higher IRCA legalization exposure is associated with changes in the recorded occupational composition of admissions during 1989–1992.

The main finding is straightforward:

1. A decline in the recorded blue-collar share appears only when the denominator excludes "no occupation.
2. The headline blue-collar share decline is largely a denominator and reporting result, not clear evidence of a collapse in blue-collar admissions.

This is therefore best understood as a data quality and KPI interpretation case study, not a causal evaluation of IRCA. The report shows how denominator choice, missingness, and administrative reporting shifts can materially alter a commonly used composition metric.

Why This Matters for a Data Analyst

This project is designed as a practical analytics exercise, not just a policy-history analysis.

A reporting or operations analytics team could easily monitor a metric like "blue-collar share" and conclude that composition changed sharply during a legalization surge. But that interpretation can be wrong if the metric depends on an unstable denominator.

The broader analyst lesson is:

> When administrative data quality or coding coverage shifts, percentage-based KPIs can move for mechanical reasons rather than substantive ones.

That makes this a useful portfolio example in:

- administrative data analysis
- dashboard KPI design
- denominator sensitivity auditing
- robustness analysis under imperfect measurement

Research Question

The analytical question is:

Do origin countries with higher IRCA legalization exposure show a true decline in blue-collar admissions share during 1989–1992, or does the result mainly reflect denominator and reporting changes?

This framing is more useful than simply asking whether IRCA "reduced" blue-collar shares, because the key empirical issue in this dataset is how the occupational denominator is constructed.

Data and Exposure Design

The analysis uses author-compiled spreadsheets based on publicly available INS Statistical Yearbooks. The report uses four harmonized country–year inputs:

- occupation
- class of admission
- gender/age
- adjustments versus new arrivals

The sample covers 19 Latin American origin countries from 1982 to 2000.

Exposure measure

Country-level IRCA exposure is defined as:

$$\text{Exposure} = \frac{\text{IRCA legalizations (1989–1992)}}{\text{Pre-IRCA admissions (1982–1985)}}.$$

Countries with larger legalization surges relative to their earlier admissions base are treated as more highly exposed.

Design

The core models are descriptive exposure-gradient panel regressions with:

- country fixed effects
- year fixed effects
- clustered standard errors by origin country

The estimates are not interpreted causally. Timing diagnostics later in the report show pre-period differences, so the regressions are used as descriptive summaries rather than identification-based treatment effects.

Main Result: The Headline Depends on the Denominator

The central portfolio result is the same-sample comparison below. It asks whether the blue-collar share result survives when the denominator is expanded to include records with no occupation.

Table 1: Main result: blue-collar share using net versus full denominator on the same sample.

	Blue share (Net denom; same sample)	Blue share (Full denom; same sample)
IRCA intensity × Post	−0.036***	−0.005
	(0.007)	(0.005)
Adjustment share	0.169*	0.127**
	(0.085)	(0.059)
Num.Obs.	351	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Same-sample comparison: rows require both blue_share and blue_fullden observed. Portfolio takeaway: denominator sensitivity suggests reporting/missingness mechanics are central.

This is the most important table in the report.

Using the net denominator (known occupations only), higher IRCA exposure is associated with a decline in the blue-collar share during 1989–1992. But when the full denominator includes "no occupation," the association becomes much smaller.

That contrast strongly suggests that the headline result is sensitive to occupation coverage and denominator construction. In other words, the apparent blue-collar share decline is not enough by itself to conclude that blue-collar admissions truly collapsed.

Mechanism: Numerator vs Denominator Growth

A falling share can reflect either:

- a declining numerator, or
- a denominator that grows faster than the numerator

To distinguish between those explanations, the project decomposes the result into blue-collar counts and denominator counts.

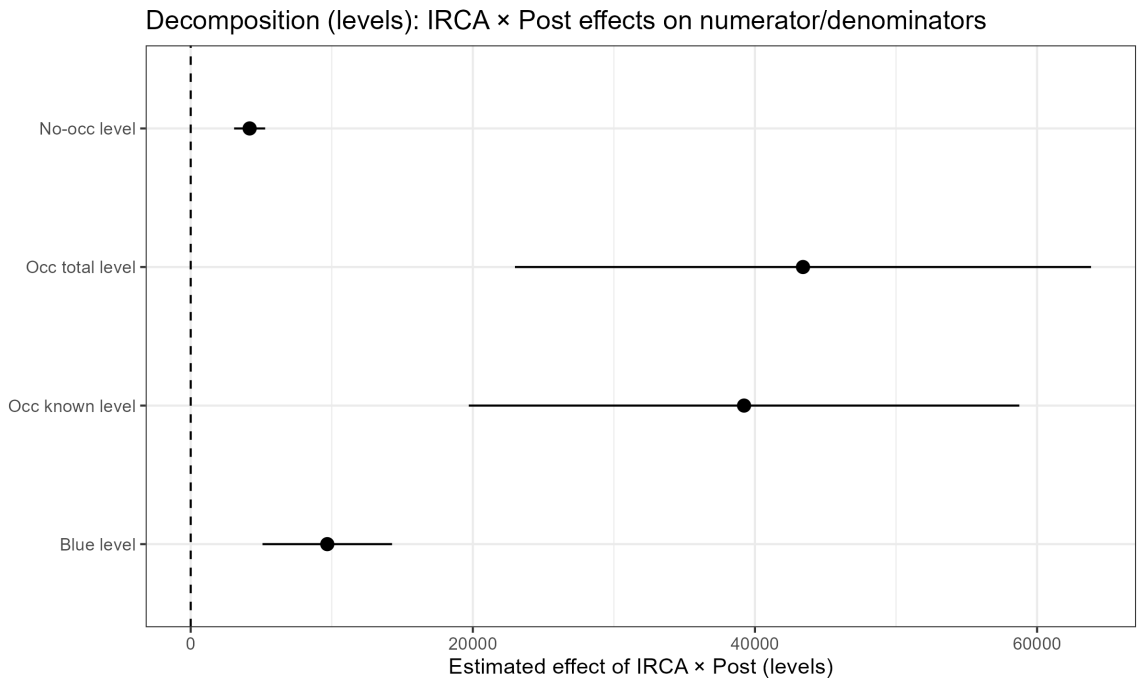

Figure 1: Decomposition in levels: IRCA × Post associations on blue-collar counts and denominators.

The decomposition supports a denominator-based interpretation. The blue-collar numerator does not show evidence of a simple collapse. Instead, the denominator grows strongly during high-exposure surge years, which is consistent with the idea that the observed share decline is largely mechanical.

That is the key analytical move in the project: not stopping at the share, but asking what changed underneath it.

Denominator Channel Evidence: "No Occupation" Share

A natural way to test whether denominator mechanics matter is to examine the behavior of the "no occupation" category itself.

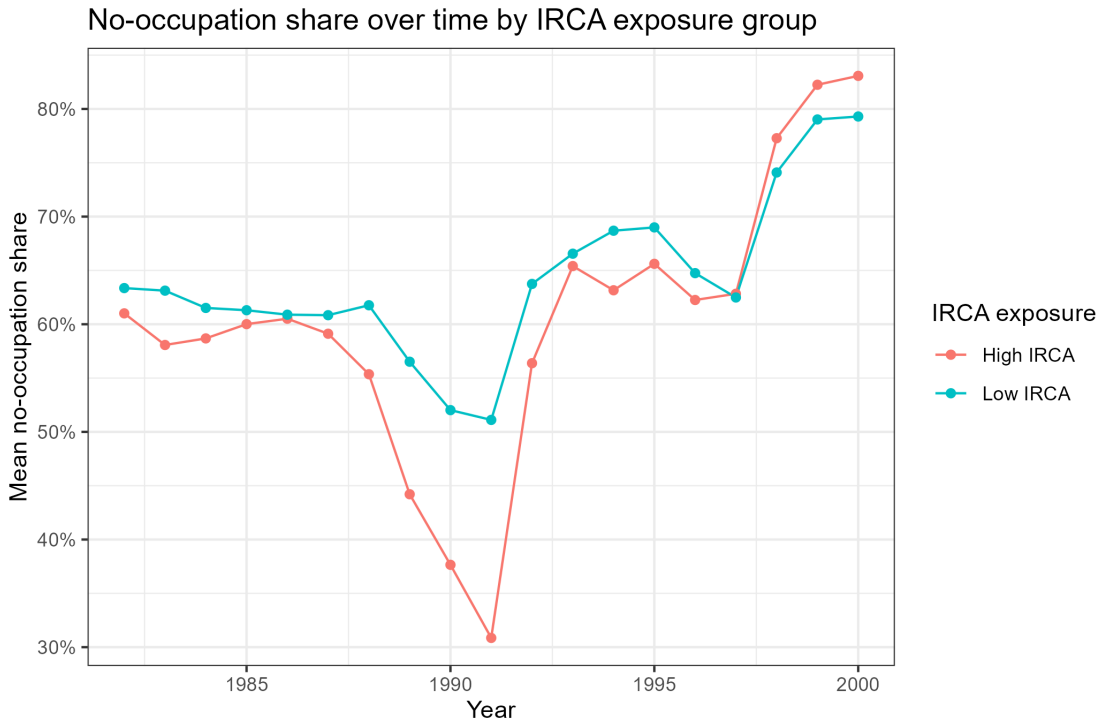

Figure 2: No-occupation share over time by IRCA exposure group.

The figure shows that more highly exposed origin countries experienced a larger rise in the no-occupation share around the IRCA surge years. That reinforces the interpretation that occupation coverage changed at the same time the net-denominator blue-collar share was falling.

For an analytics audience, this is the main operational takeaway: the blue-collar KPI should not be interpreted in isolation. It should be monitored jointly with the full-denominator share and the no-occupation share.

Robustness: Is the Result Driven by One Country?

A common concern in small-country panels is that one country may dominate the estimated relationship. The leave-one-country-out exercise checks that.

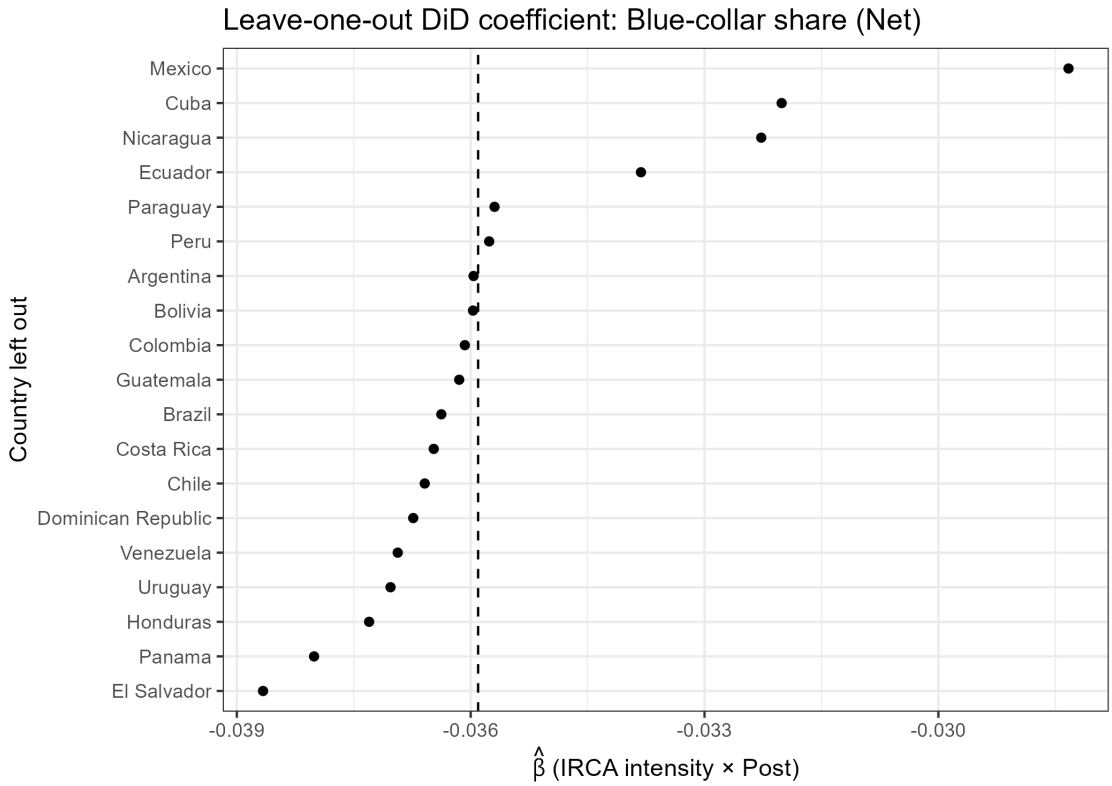

Figure 3: Leave-one-country-out influence on the baseline net-denominator coefficient.

The result is not driven by any single origin country. The coefficient moves as countries are removed one at a time, but the overall pattern remains intact.

This matters because it shows that the denominator-sensitivity story is not just a Mexico artifact or a single-country anomaly.

Small-Cluster Inference

Because the panel has only 19 origin-country clusters, conventional clustered standard errors may be fragile. To address that, the project adds a wild cluster bootstrap check for the baseline net-denominator specification.

Table 2: Wild cluster bootstrap inference for the baseline net-denominator specification.

Model	Parameter	Bootstrap p-value	Bootstrap t-stat
lm with country+year FE (dummies)	IRCA intensity × Post	0.03361	-5.154

The bootstrap is estimated using an equivalent country-and-year dummy regression because of package compatibility constraints. It serves as an inference check rather than a separate substantive model.

Magnitudes

To make the results more interpretable, the project reports standardized magnitudes under both denominator definitions.

Table 3: Standardized effect sizes for blue-collar share using the net denominator.

Magnitude	Effect (pp / %)
Per 1 unit exposure	-3.59
Per 1 SD exposure	-7.61
Per IQR exposure	-2.46
As % of mean(Y): 1 SD	-22.22
As % of mean(Y): IQR	-7.18

Table 4: Standardized effect sizes for blue-collar share using the full denominator.

Magnitude	Effect (pp / %)
Per 1 unit exposure	-0.54
Per 1 SD exposure	-1.13
Per IQR exposure	-0.37
As % of mean(Y): 1 SD	-8.63
As % of mean(Y): IQR	-2.79

The contrast in magnitudes reinforces the same conclusion as the main result table: the apparent blue-collar decline is materially smaller once denominator coverage is handled more conservatively.

Reporting Recommendation for Analysts

For dashboarding or operations monitoring, the report recommends presenting three metrics together:

1. Blue share (net denominator)
2. Blue share (full denominator)
3. No-occupation share

That combination makes it easier to tell whether a change in the headline composition metric reflects:

- genuine compositional change, or
- reporting/coding shifts in the underlying records

In practical terms, this is the best stakeholder-facing takeaway of the project.

Interpretation and Limits

This is a descriptive exposure-gradient analysis, not a causal design.

That distinction matters because the event-study timing checks show pre-period differences. More highly exposed origin countries were already on somewhat different trajectories before the IRCA surge, so the interaction coefficients should be interpreted as summary associations rather than causal treatment effects.

That limitation does not weaken the portfolio value of the project. In fact, it strengthens it: the report shows that the analyst did not confuse statistical output with identification.

Main Takeaways

This project produces four main takeaways:

1. The headline blue-collar share result is highly sensitive to denominator choice.
2. The underlying mechanism is better described as denominator/reporting expansion than as a pure decline in blue-collar counts.
3. The no-occupation category is central to interpreting occupational composition during surge years.
4. Administrative-data KPIs should be audited for denominator instability before being used for substantive conclusions.

Portfolio Takeaway

As a portfolio project, this case study demonstrates the ability to:

- build a reproducible panel from raw administrative-style spreadsheets
- define and test a country-level exposure measure
- identify denominator sensitivity in a commonly used KPI
- decompose a share into numerator and denominator channels
- run and interpret robustness checks without overstating causality
- translate a technical result into a stakeholder-facing measurement lesson

That is the main reason this project works well as data analyst portfolio material: it shows judgment, not just model execution.

How to Run

1. Put the raw Excel inputs in `data_raw/`
2. Open the project file
3. Run `Rscript run.R` or source `run.R`
4. Render `paper.Rmd`

Outputs are written to:

- `outputs/tables/`

- `outputs/figures/`
- `outputs/logs/`
- `data_clean/panel.rds`

Sources

- U.S. Immigration and Naturalization Service (INS). *Statistical Yearbook of the Immigration and Naturalization Service*, various years.

Appendix

Appendix A: Portfolio dashboard

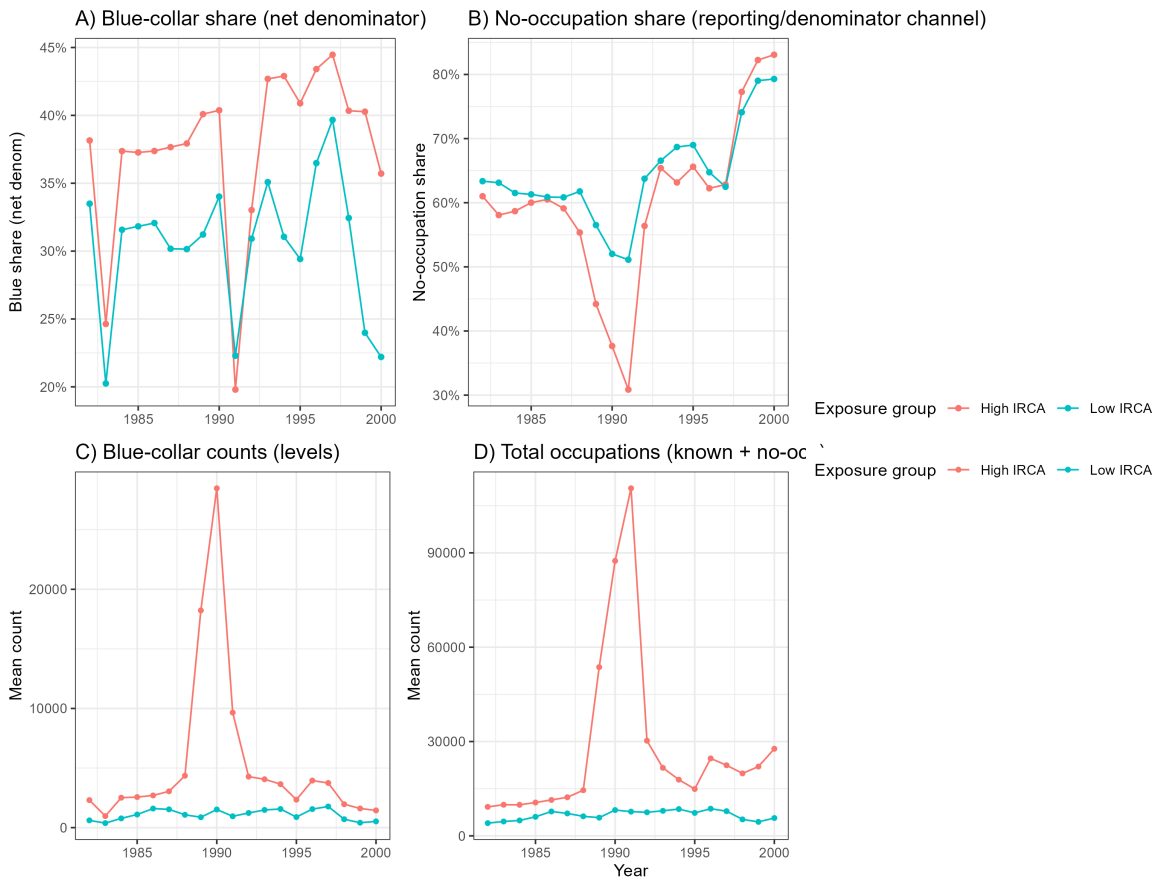

Figure 4: Dashboard of shares, no-occupation share, and counts by IRCA exposure group.

Appendix B: Data quality and coverage

Table 5: Missingness by variable at the country–year level.

Variable	Obs.	Missing	Total	Share missing (%)
Male share	133	228	361	63.2
Family share	352	9	361	2.5
Employment share	352	9	361	2.5
Humanitarian share	352	9	361	2.5
Other share	352	9	361	2.5
Adjustment share	352	9	361	2.5
Blue share	353	8	361	2.2
Agriculture share	353	8	361	2.2
Manual share	353	8	361	2.2
Services share	353	8	361	2.2
White share	353	8	361	2.2
No-occupation share	353	8	361	2.2

Table 6: Coverage by origin country for key variables.

Country	Years total	Years covered (key vars)	Coverage ratio
Argentina	19	19	1.000
Bolivia	19	19	1.000
Brazil	19	19	1.000
Chile	19	19	1.000
Colombia	19	19	1.000
Costa Rica	19	19	1.000
Cuba	19	19	1.000
Dominican Republic	19	19	1.000
Ecuador	19	19	1.000
El Salvador	19	19	1.000
Guatemala	19	19	1.000
Honduras	19	19	1.000
Mexico	19	19	1.000
Nicaragua	19	19	1.000
Panama	19	19	1.000
Peru	19	19	1.000
Venezuela	19	19	1.000
Uruguay	19	15	0.789
Paraguay	19	13	0.684

Appendix C: Baseline regression panels

Table 7: Baseline TWFE (Total): occupation shares.

	Blue	Agri	Manual	Services	White
IRCA intensity × Post	−0.025***	0.029***	0.004	−0.007**	0.004
	(0.005)	(0.004)	(0.005)	(0.003)	(0.005)
Num.Obs.	353	353	353	353	353
Mean of Y (share units)	0.342	0.050	0.392	0.224	0.384

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp changes per one-unit exposure. Total specs exclude adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 8: Baseline TWFE (Total): no-occupation share and class-of-admission shares.

	No-Occ	Family	Employment	Humanitarian	Other
IRCA intensity × Post	−0.035***	−0.052***	−0.021*	−0.004***	−0.002
	(0.007)	(0.004)	(0.011)	(0.001)	(0.001)
Num.Obs.	353	352	352	352	352
Mean of Y (share units)	0.620	0.542	0.301	0.067	0.013

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp changes per one-unit exposure. Total specs exclude adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 9: Baseline TWFE (Net): occupation shares controlling for adjustment share.

	Blue	Agri	Manual	Services	White
IRCA intensity × Post	−0.036***	0.029***	−0.007	−0.006*	0.013*
	(0.007)	(0.004)	(0.006)	(0.003)	(0.006)
Adjustment share	0.169*	−0.006	0.164*	−0.019	−0.144
	(0.085)	(0.027)	(0.082)	(0.032)	(0.093)
Num.Obs.	351	351	351	351	351
Mean of Y (share units)	0.342	0.050	0.392	0.224	0.384

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp per one-unit exposure. Net specs include adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 10: Baseline TWFE (Net): no-occupation share and class-of-admission shares controlling for adjustment share.

	No-Occ	Family	Employment	Humanitarian	Other
IRCA intensity × Post	−0.025***	−0.027**	−0.008	−0.016**	−0.009
	(0.005)	(0.011)	(0.011)	(0.006)	(0.009)
Adjustment share	−0.149*	−0.400**	−0.206	0.192**	0.108
	(0.081)	(0.152)	(0.128)	(0.091)	(0.138)
Num.Obs.	351	352	352	352	352
Mean of Y (share units)	0.619	0.542	0.301	0.067	0.013

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp per one-unit exposure. Net specs include adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Appendix D: Additional robustness

Table 11: Specification comparison for blue-collar share (net denominator).

	Baseline Net	Country trends	WLS admissions	Balanced Net
IRCA × Post	−0.036***	−0.036***	−0.036***	−0.037***
	(0.007)	(0.008)	(0.003)	(0.007)
Adjustment share	0.169*	0.160	0.150**	0.190*
	(0.085)	(0.099)	(0.069)	(0.090)
Num.Obs.	351	351	351	323

* p <0.1, ** p <0.05, *** p <0.01
Comparison of key robustness specifications for blue-collar share.

Table 12: Robustness to origin-specific linear trends.

	Baseline Total	Baseline Net	Trends Total	Trends Net
IRCA intensity × Post	−0.025***	−0.036***	−0.025***	−0.036***
	(0.005)	(0.007)	(0.005)	(0.008)
Adjustment share		0.169*		0.160
		(0.085)		(0.099)
Num.Obs.	353	351	353	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Trends specs include origin-specific linear trends (country[t]).

Table 13: Additional sensitivity checks for blue-collar share.

	Baseline Net	Drop 1995–1997	Drop UY+PY	Window 1982–1995
IRCA × Post	−0.036***	−0.042***	−0.037***	−0.037***
	(0.007)	(0.008)	(0.007)	(0.008)
Adjustment share	0.169*	0.224**	0.190*	0.182*
	(0.085)	(0.092)	(0.090)	(0.092)
Num.Obs.	351	300	323	260

* p <0.1, ** p <0.05, *** p <0.01
All models: Country & Year FE; SE clustered by country. Outcome: blue_share.

Table 14: Weighted specifications.

	Net (unweighted)	Net (w: admissions)	Net (w: known occ.)
IRCA intensity × Post	−0.036***	−0.036***	−0.034***
	(0.007)	(0.003)	(0.004)
Adjustment share	0.169*	0.150**	0.105
	(0.085)	(0.069)	(0.084)
Num.Obs.	351	351	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country.

Table 15: Balanced-panel restriction.

	Blue Net (balanced)
IRCA × Post	−0.037***
	(0.007)
Adjustment share	0.190*
	(0.090)
Num.Obs.	323

* p <0.1, ** p <0.05, *** p <0.01
Balanced panel: keeps only countries observed in all years of the Net sample.

Table 16: Alternative exposure construction and Mexico exclusion.

	Blue Net (baseline)	Blue Net (alt denom)	Blue Net (–MX)	Blue Net (alt –MX)
IRCA intensity × Post	−0.036***		−0.028**	
	(0.007)		(0.013)	
IRCA intensity (alt) × Post		−0.009***		−0.007**
		(0.002)		(0.003)
Adjustment share	0.169*	0.169*	0.163	0.163
	(0.085)	(0.085)	(0.106)	(0.106)
Num.Obs.	351	351	332	332

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Outcome is share in [0,1]; coefficients are pp.

Appendix E: Decomposition tables and log specification

Table 17: Decomposition in levels.

	Blue level	Occ known level	Occ total level	No-occ level
IRCA × Post	9682.575***	39 226.652***	43 404.916***	4178.264***
	(2342.817)	(9956.342)	(10 418.047)	(561.725)
Adjustment share	−973.413	−18 202.159	−204.179	17 997.980
	(5699.283)	(21 227.495)	(32 984.662)	(17 559.005)
Num.Obs.	352	352	352	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Levels are counts.

Table 18: Decomposition in log(1+x).

	ln(1+Blue)	ln(1+Occ known)	ln(1+Occ total)	ln(1+No-occ)
IRCA × Post	0.098*	0.192***	0.131***	0.056
	(0.051)	(0.037)	(0.036)	(0.034)
Adjustment share	1.092	0.776	0.608	0.224
	(0.636)	(0.469)	(0.470)	(0.513)
Num.Obs.	352	352	352	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Log specs use ln(1+x).

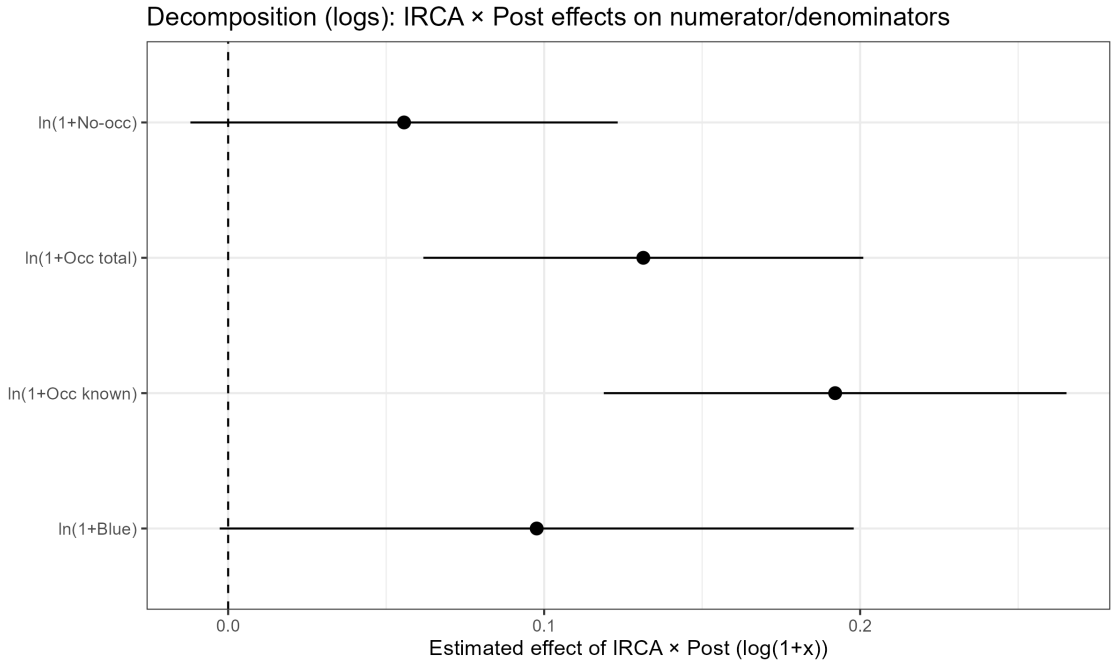

Figure 5: Decomposition in log(1+x): proportional associations on numerator and denominators.

Appendix F: Timing diagnostics

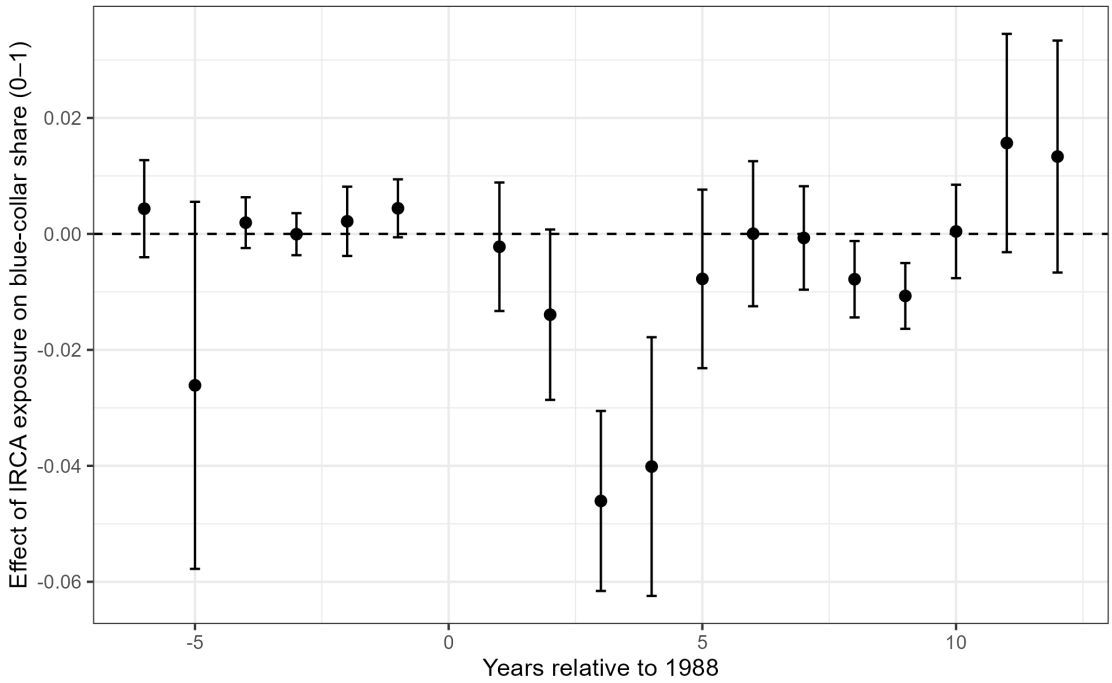

Figure 6: Event-study: exposure-by-year associations for blue-collar share (relative to 1988).

Table 19: Wald test of joint significance for pre-1988 event-study coefficients.

Test	Statistic	df1	df2	p_value
Wald joint pre-trends (years < 1988)	13.54	6	298	0

The pretrend diagnostic is important for interpretation. It indicates that higher-exposure countries were already evolving differently before the IRCA surge, which is why the main estimates are presented as descriptive rather than causal.

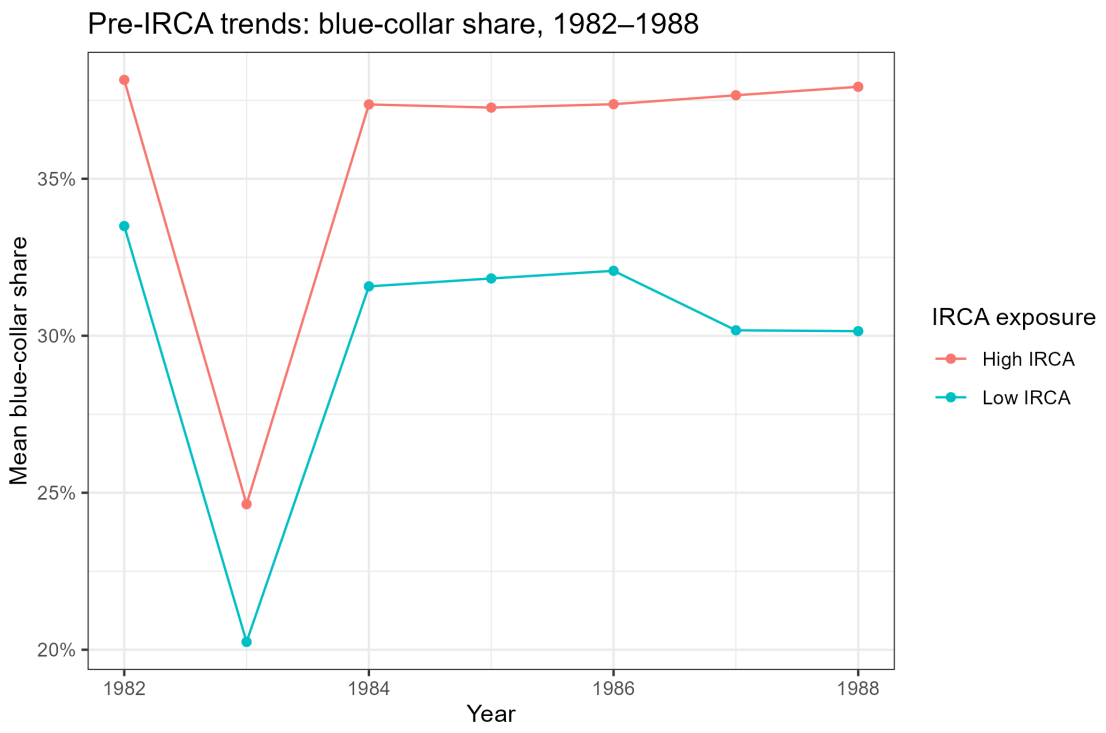

Figure 7: Pre-IRCA group means for blue-collar share by exposure group, 1982–1988.

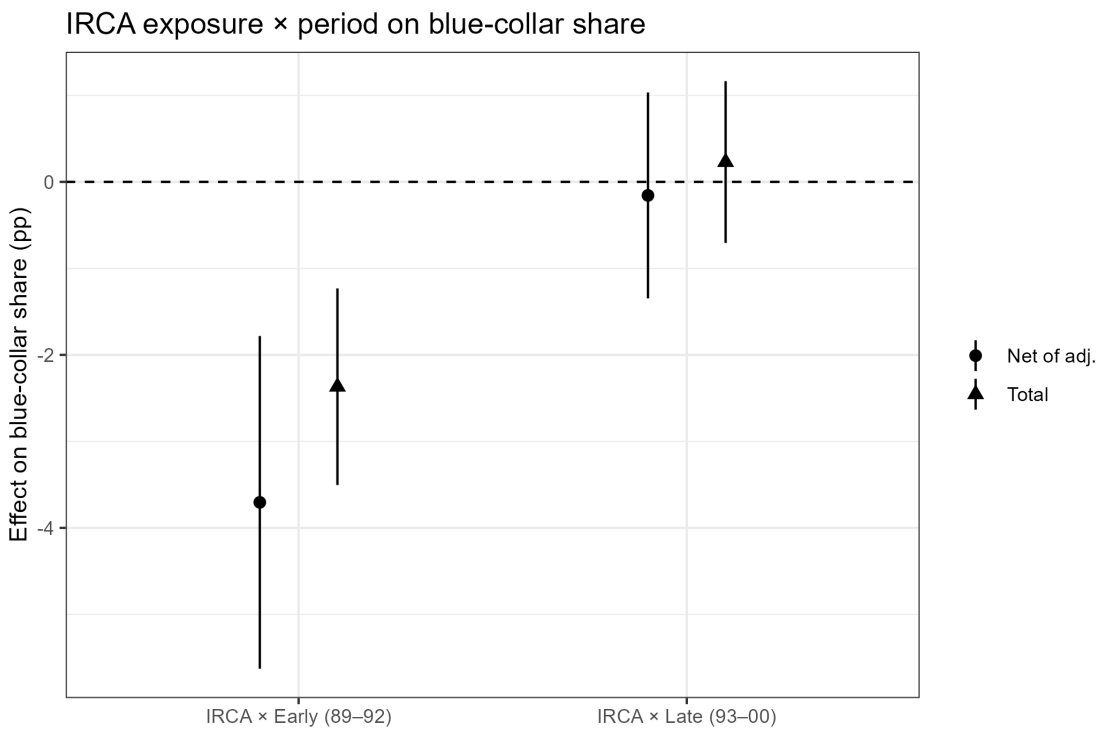

Figure 8: Early versus late post-period exposure-gradient associations for blue-collar share.

Appendix G: Additional exploratory outputs

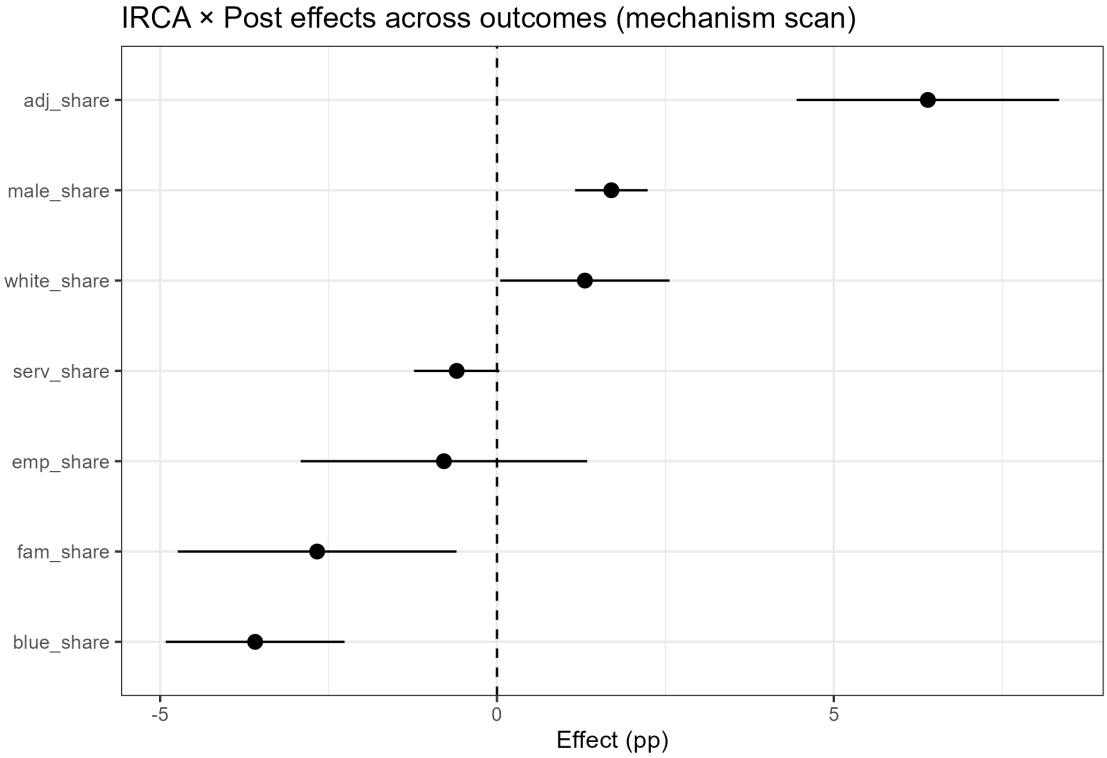

Figure 9: Exposure × Post associations across outcomes.

Appendix H: QA artifact

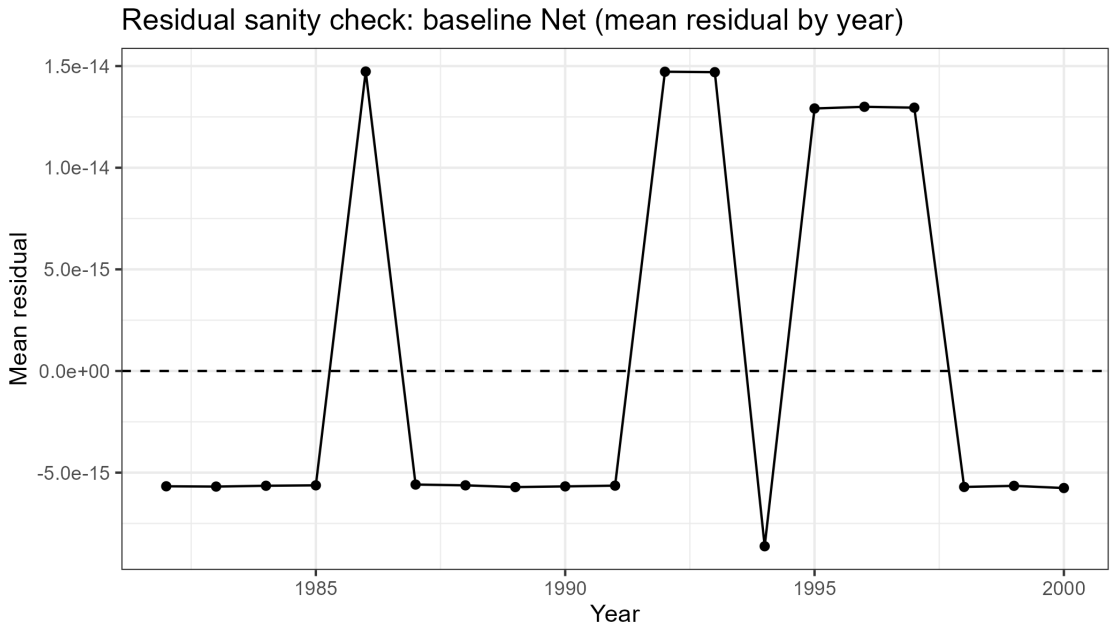

Figure 10: Mean residual by year in the baseline net-denominator specification.